Presentation Title

Optional Subtitle

Author Name

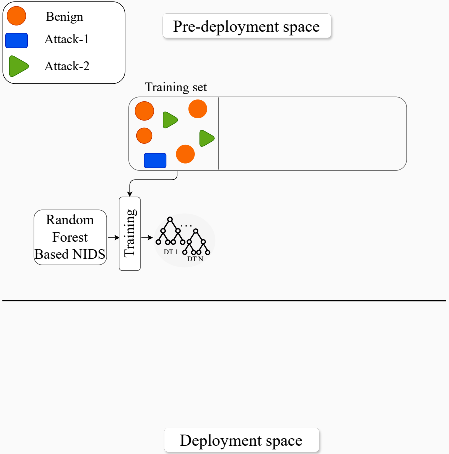
Affiliation

NOMS 2026

Introduction

ML-based NIDS Workflow: Train → Validate → Test → Deploy

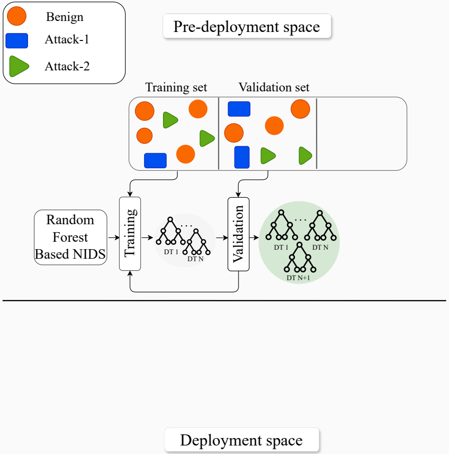
- The model is trained on a training set $\mathcal{D}_{\textbf{train}}$.



ML-based NIDS Workflow: Train $\rightarrow$ Validate $\rightarrow$ Test $\rightarrow$ Deploy

- The model is trained on a training set $\mathcal{D}_{\textbf{train}}$.
- A separate validation split $\mathcal{D}_{\textbf{val}}$ to refine the model hyperparameters.

Hyperparameters: settings chosen before training that influence model behavior.
Example: In RF, too few trees can give unstable predictions, while too many can increase computation and overfit the training data.



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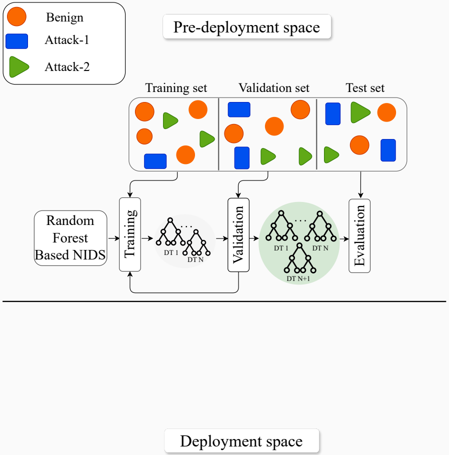
- A final test set, $\mathcal{D}_{\textsf{test}}$, measures performance on unseen data.



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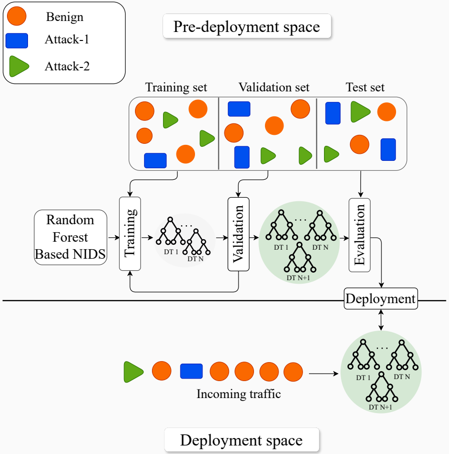
- The selected model is deployed to monitor live traffic.



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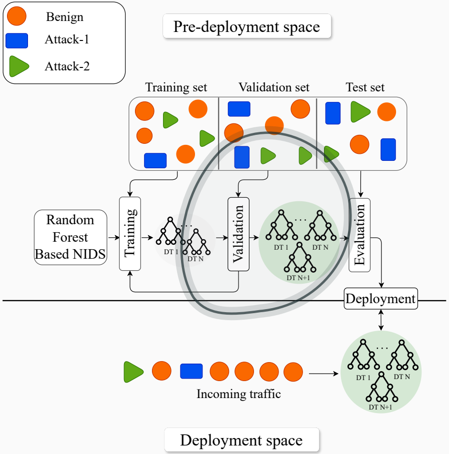

$\Longrightarrow$ This work focuses on the **configuration step** and its impact on **NIDS performance**.

Background

Method

Results

Conclusion

Questions?